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Poverty mapping in the age of machine learning[☆]Paul Corral^{a,*}, Heath Henderson^b, Sandra Segovia^a^a The World Bank Group - Poverty and Equity Global Practice, United States of America^b Zimbleman College of Business, Drake University, United States of America

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ABSTRACT

Recent years have witnessed considerable methodological advances in poverty mapping, much of which has focused on the application of modern machine-learning approaches to remotely-sensed data. Poverty maps produced with these methods generally share a common validation procedure, which assesses model performance by comparing sub-national poverty estimates with survey-based, direct estimates. While unbiased, direct estimates can be imprecise measures of true poverty rates, meaning that it is unclear whether these validation procedures are informative of actual model performance. In this paper, we use a rich dataset from Mexico to provide a more rigorous assessment of the modern approach to poverty mapping by evaluating its performance against a credible ground truth. We find that the modern method under-performs relative to benchmark traditional methods, largely because of the limited predictive capacity of remotely-sensed covariates. For a given covariate set, we also find that machine learning produces more biased poverty estimates than the traditional procedures, particularly for the poorest geographic areas.

1. Introduction

Poverty maps provide granular estimates of poverty at the sub-national level in order to better inform the targeting of resources and support the design of interventions tailored to local needs (Bedi et al., 2007; Elbers et al., 2007). As household surveys are generally unrepresentative for small geographic areas, producing detailed poverty maps typically requires using small-area estimation, which consists of using statistical methods to predict poverty levels on the basis of household- or area-level characteristics. The basic procedure entails two steps: (1) using household survey data, fit a statistical model of some welfare measure (e.g., poverty, income, or wealth) to household- or area-level characteristics, and (2) apply the model to data available for all households or geographic units to estimate poverty at the sub-national level. At a minimum, the procedure then requires a household survey and data that covers all geographic areas to produce a sufficiently detailed map.

The literature on small-area estimation is rich and several variations on this basic procedure have been proposed. Unit-level models generally use survey and census data to conduct estimation and prediction at the household level, assuming a parametric relationship between the welfare measure and covariates (Hentschel et al., 1998; Elbers et al.,

2003; Molina and Rao, 2010). Unit-level models are not well-suited to situations where the survey and census correspond to different years, which is often the case in developing countries where censuses are conducted infrequently. Area-level models are the leading parametric alternative for off-census years and only use aggregate data on the geographic units of interest for estimation and prediction (Fay and Herriot, 1979; Torabi and Rao, 2014).

Recent developments in poverty mapping have focused on the application of machine learning to remotely-sensed data, largely in response to the issue of out-of-date census information. These approaches also use survey-derived welfare measures, but are distinguished by a first stage where a machine-learning model is fit to remotely-sensed covariates (e.g., data derived from satellite imagery or call detail records) rather than census-based covariates. For example, Chi et al. (2022) matched data from several remotely-sensed data sources to survey-based measures of “village” wealth for 56 countries, and then fit a prediction model using gradient boosting machines.¹ They then applied the model to the populated surface of 135 low- and middle-income countries to develop granular poverty maps for each country. Similar approaches have been used for individual countries, including Rwanda (Blumenstock et al., 2015), Bangladesh (Steele et al., 2017),

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¹ Their remotely-sensed data includes high-resolution satellite imagery, mobile phone data, topographic maps, and connectivity data from Facebook.

Belize (Hersh et al., 2021), and Sri Lanka (Engstrom et al., 2022), among others.²

This alternative method – which we refer to as the “modern” approach to poverty mapping – consists of several practices that differ from more traditional approaches. Most obviously, the modern approach substitutes non-parametric statistical methods for the parametric approaches traditionally used for poverty mapping. In addition, the modern approach relies heavily on remotely-sensed covariates rather than covariates derived from census or administrative data.³ Further, maps produced with machine-learning methods generally share a common validation procedure. The key feature of this procedure is that model performance is assessed by calculating the R^2 from a regression of the observed poverty measures on the estimates for the same geographic units.⁴ There are several limitations to this procedure, but the most critical is that it typically uses direct, sample-based estimates rather than true poverty measures as a performance benchmark. While direct estimates are unbiased, they can be imprecise estimates of the true measures, meaning that it is unclear whether the validation procedure is informative of actual model performance (see Section 3).

In this paper, we provide a more rigorous assessment of the performance of the modern approach to poverty mapping. Our approach consists of constructing a pseudo-census (hereafter “census”) that we use to conduct several simulation experiments. Our experiments entail repeatedly drawing survey samples from the census where each sample produces direct poverty estimates for the sampled geographic areas. With each sample we obtain small-area estimates using traditional and modern mapping methods. The presence of the census means that we observe the “true” poverty measure corresponding to the direct estimate. We use these true poverty measures to gain insight into the performance of machine-learning methods when validated against a credible ground truth. As population censuses rarely collect detailed data on income or expenditures, we construct our census from a large-scale survey conducted in Mexico: the Mexican Intercensal Survey of 2015.

The results of our simulations can be summarized as follows. First, we find that poverty maps produced using machine learning and remotely-sensed covariates tend to perform poorly relative to the traditional approach, both in terms of the mean squared error of the estimates and simulated targeting performance. Second, we find that this under-performance is due to the low predictive capacity of the remotely-sensed covariates. Machine learning performs comparably to, and sometimes better than, the traditional methods when both methods use the same covariates. Third, we show that the performance advantage of machine learning for a given covariate set results from the relatively low variance of its estimates, but this variance reduction comes at the cost of increasing the bias of the estimates, particularly for the poorest areas. This bias–variance trade-off is especially stark when using remotely-sensed covariates and ultimately raises practical questions about the use and production of modern poverty maps.

Our paper builds on the work of Corral et al. (2021a) and Corral et al. (2022), who similarly used the Mexican Intercensal Survey to examine the performance of several mapping methods. Corral et al. (2021a) focused exclusively on traditional mapping approaches and did

not consider the performance of machine-learning methods. While Corral et al. (2022) consider the performance of machine learning, their implementation only uses census-based covariates and thus does not examine the performance of the standard implementation that relies on remotely-sensed covariates. Importantly, we are not the first paper to evaluate machine-learning methods relative to independent census data. While Yeh et al. (2020) and Chi et al. (2022) validate their models relative to a census, they focus on wealth rather than poverty and do not address all the limitations of R^2 as a performance metric (see Section 3). Finally, van der Weide et al. (2022) examine the performance of machine-learning methods combined with remotely-sensed data, but not relative to true poverty measures.

This paper then represents the first attempt to rigorously assess the performance of modern poverty-mapping methods relative to a ground truth. The scope of our analysis is nevertheless limited in some important ways. For one, we confine our attention to the performance of machine learning for predicting income poverty and do not consider other welfare measures, such as wealth or multi-dimensional poverty. We also focus exclusively on the case of Mexico because of the unique data requirements of our experiments and therefore do not assess the external validity of our findings. A final limitation of our analysis is that we do not consider all possible uses of machine learning for poverty mapping. For example, we do not consider the issue of “feature extraction”, which is a data-processing technique that builds the covariates for mapping exercises by applying machine learning to unstructured input data (e.g., satellite imagery).⁵

In what follows, Section 2 discusses the various methods we examine, including machine-learning methods and the traditional approaches we use as performance benchmarks. Section 3 considers the issue of model validation by presenting a critical evaluation of R^2 as a validation metric and proposing some alternative measures. Section 4 then discusses the data we use for our simulation experiments and Section 5 presents the results. Finally, Section 6 provides concluding remarks, including practical recommendations for the production and use of poverty maps.

2. Methods

This section describes the methods we use in our poverty-mapping simulations. The first subsection discusses machine learning, with a special focus on gradient boosting, as it is among the most popular machine-learning methods for poverty mapping. The second subsection then discusses more traditional approaches to poverty mapping, which we use as the benchmark in our performance assessment of the machine-learning methods.

2.1. Machine learning

The goal of machine learning is to develop high-performance algorithms for prediction, classification, and clustering/grouping tasks (Varian, 2014; Athey, 2018; Athey and Imbens, 2019). Such tasks can be divided into supervised and unsupervised learning problems. While unsupervised learning focuses on identifying clusters of observations that are similar in terms of their features, supervised learning uses a set of features to predict some outcome of interest. Supervised learning can be further divided into regression and classification tasks, where regression is concerned with predicting continuous outcomes and classification focuses on categorical outcomes. While there are many machine-learning methods used for regression and classification

² See Jean et al. (2016), Pokhriyal and Jacques (2017), Yeh et al. (2020), Lee and Braithwaite (2020), Aiken et al. (2022) and Newhouse et al. (2022) for examples of other recent applications.

³ Traditional area-level models occasionally use remotely-sensed data. See Seitz (2019) for an application using publicly-available geospatial data.

⁴ For examples of the above procedure, see Jean et al. (2016), Pokhriyal and Jacques (2017), Steele et al. (2017), Lee and Braithwaite (2020), Yeh et al. (2020), or Chi et al. (2022). Note that a few studies use a similar procedure, but fit the machine-learning model to household-level data and then aggregate the model's predictions to the geographic unit of interest (Blumenstock et al., 2015; Hersh et al., 2021; Aiken et al., 2022).

⁵ Smythe and Blumenstock (2022), for example, used a convolutional neural network to extract features from high-resolution satellite imagery in Nigeria and then used those features to create a poverty map using gradient boosting. See Jean et al. (2016) or Chi et al. (2022) for other applications of a similar procedure.

tasks (e.g., lasso, random forests, or support vector machines), the following presents some key ideas by focusing on one of the leading methods: gradient boosting.

Originally developed by Friedman (2001), gradient boosting machines are a family of machine-learning techniques that combine a large number of weak learners to form a stronger ensemble prediction. Unlike some common ensemble techniques (e.g., random forests) that simply average models in the ensemble, gradient boosting adds models to the ensemble sequentially by fitting new weak learners to the negative gradient of the chosen loss function. With the classic squared-error loss function, this amounts to sequentially fitting new models to the current residuals of the ensemble.⁶ Extreme gradient boosting (XGBoost) is a particularly popular implementation of gradient boosting that uses classification and regression trees as the base models or weak learners. The popularity of XGBoost is due to the fact that it is fast, scalable, and has been shown to outperform competing methods across a wide range of problems (Chen and Guestrin, 2016).

Classification and regression trees are based on a sequential binary splitting of the covariate space that serves to partition the data into subsamples that are increasingly homogeneous in terms of the outcome variable. A tree begins with a single root node that is split into two child nodes according to some decision rule. A decision rule pertains to a single explanatory variable and observations are assigned to the child nodes depending on whether they meet some condition associated with the explanatory variable.⁷ The child nodes can be further split into new child nodes based on different decision rules that involve other explanatory variables and cutoff points. Any child node that is not further split is referred to as a terminal node or leaf, and each leaf is associated with a parameter that provides a prediction for the subsample associated with that leaf. For any given observation, the ensemble prediction used by XGBoost is the sum of that observation's predictions across all trees in the ensemble.

To fix ideas, let y_i^d denote the outcome of interest (e.g., direct estimates of poverty rates) for observation or entity i , and let x_i denote a vector of explanatory variables. In line with the above, the predicted value of the outcome y_i^p is given by the sum of predictions across the individual trees in the ensemble:

$$y_i^p = \sum_t f_t(x_i) \quad (1)$$

where the trees are indexed by t and $f_t(x_i)$ gives the prediction of a given tree as a function of x_i . To build the trees used in the model, XGBoost greedily minimizes the following regularized objective function:

$$\mathcal{L} = \sum_i l(y_i^d, y_i^p) + \sum_t r(f_t) \quad (2)$$

where $l(y_i^d, y_i^p)$ is the loss associated with the i th observation and $r(f_t)$ is a regularization term that serves to mitigate overfitting.⁸ In our implementation, we use the classic squared-error loss such that $l(y_i^d, y_i^p) = (y_i^d - y_i^p)^2$.

XGBoost uses the following specification of the regularization term:

$$r(f_t) = \gamma M_t + \frac{1}{2} \lambda \sum_j m_{tj}^2 \quad (3)$$

where M_t is the total number of leaves in tree t , m_{tj} is the prediction associated with the j th leaf in a given tree, and γ and λ are hyperparameters that must be chosen by the user. The regularization term serves to

mitigate overfitting by penalizing complicated trees and smoothing the predictions associated with each tree. That is, the method relies on a large number of weak learners and the regularization term helps control the complexity of the learners. It is important to note that the above specification of the regularization term is only one of many possible specifications, though Chen and Guestrin (2016) claim that it is simpler than the leading alternatives and easier to parallelize. Further note that when the hyperparameters are set to zero, the objective function becomes equivalent to that used in traditional gradient boosting.

XGBoost requires the specification of several hyperparameters that must be chosen by the user, including the hyperparameters in the regularization term, the maximum tree depth, and the number of trees in the ensemble. There are two additional hyperparameters that can be used to further prevent overfitting. First, one can use column or feature subsampling where each tree in the ensemble is built based on a (uniformly) random subset of all explanatory variables. This procedure can also speed up the algorithm because it reduces the number of covariates that must be searched across when creating new splits. Second, one can use shrinkage by scaling the predictions associated with each tree by a factor η , which is often called the learning rate. The predicted value of the outcome at iteration t is then given by the sum of the previous prediction and the current prediction scaled by the learning rate. Shrinkage reduces the influence of each tree so that no individual tree has too much influence on the ensemble prediction.

One possible approach to hyperparameter selection is to use a grid search where the user (1) specifies the domain of the hyperparameters, (2) applies the model to every combination of hyperparameters, and (3) selects the hyperparameters that perform the best in terms of some metric (e.g., mean squared prediction error in the validation dataset). Grid search, however, is computationally inefficient because hyperparameter proposals are uninformed by previous evaluations. We instead use a Bayesian hyperparameter optimization procedure, which sequentially updates a probability model that relates the hyperparameter space to the performance metric and then uses the model to intelligently explore the hyperparameter space (Bergstra et al., 2011). Specifically, our implementation of gradient boosting relies on the open-source library XGBoost, which we combine with the Hyperopt library that conducts Bayesian hyperparameter optimization using the tree-structured Parzen estimator.

2.2. Traditional methods

We benchmark the performance of the machine-learning approach against two traditional poverty-mapping methods: the unit-level and area-level models. Regarding unit-level models, the well-known methods of Elbers et al. (2003) and Molina and Rao (2010) are based on the nested error model originally proposed by Battese et al. (1988). This model assumes that (transformed) equalized household income or consumption relates to a vector of household-level covariates according to the following model:

$$w_{iv} = x_{iv}\theta + \mu_i + \xi_{iv} \quad (4)$$

where w_{iv} denotes the transformed equalized income of household v in location i , x_{iv} represents a vector of household-specific characteristics, θ is a vector of coefficients on the household characteristics, μ_i is location-specific random effect such that $\mu_i \sim N(0, \sigma_\mu^2)$, and ξ_{iv} is a household-level error term such that $\xi_{iv} \sim N(0, \sigma_\xi^2)$.

The parameters of the model are typically estimated via restricted maximum likelihood or Henderson's Method III (Henderson, 1953).⁹ Once the model is estimated, the parameters are then used to repeatedly generate income estimates for all households in a corresponding census

⁶ See Natekin and Knoll (2013) for an accessible introduction to gradient boosting.

⁷ For example, for a continuous explanatory variable, one determines whether a given observation falls above or below some numeric cutoff.

⁸ For an in-depth discussion and description of the XGBoost algorithm refer to Chen and Guestrin (2016). Note that the fact that XGBoost is a greedy algorithm means that a global minimum is not guaranteed.

⁹ Henderson's Method III is coupled with generalized least squares to obtain the regression coefficients.

dataset. That is, the income level of each household is predicted K times using the model estimates as follows:

$$w_{ivk}^* = x_{iv}\hat{\theta} + \mu_{ik}^* + \xi_{ivk}^* \quad (5)$$

where k indexes replications and where μ_{ik}^* and ξ_{ivk}^* are drawn from their assumed distributions.¹⁰ As a final step, the replicates can be used to calculate K estimates of the poverty rate for each geographic entity and these estimates can then be averaged to arrive at a final prediction for each area. Note that there are multiple versions of the unit-level model, distinguished primarily through different treatments of μ_{ik}^* and ξ_{ivk}^* . We use a particular version known as the “census empirical best” method, which is a variant of the “empirical best” method of [Molina and Rao \(2010\)](#).¹¹

The key assumption of the unit-level model is that the explanatory variables in the survey and census share the same definitions and have similar distributions across the two data sources. This assumption is generally violated if the census is outdated – which is often the case in developing countries – and in this event the application of the unit-level model will lead to similarly outdated poverty maps ([Lange et al., 2018](#)). Importantly, machine-learning approaches to poverty mapping are commonly motivated by the fact that censuses are outdated and are thus viewed as a way to produce timely poverty maps in a data-constrained environment. Stated differently, given the differing data requirements between the unit-level model and the machine-learning approach, the two methods are not generally considered to be direct competitors. The unit-level model nevertheless serves as a useful performance benchmark, as it is often viewed as the “gold standard” of poverty mapping.

In off-census years, the area-level model is generally considered the leading traditional method ([Corral et al., 2022](#)). First proposed by [Fay and Herriot \(1979\)](#), area-level models conduct estimation and prediction using data aggregated to the geographic area of interest. The area-level model consists of two basic parts. The first part of the model links the actual or true poverty rates y_i^a for the i th entity to a vector of area-level covariates x_i as follows:

$$y_i^a = x_i\beta + \varepsilon_i \quad (6)$$

where β is a vector of parameters to be estimated and ε_i represents random area-level effects that are assumed to be mean zero with a constant variance σ_ε^2 .¹² Since we lack information on the true poverty rates, this model cannot be immediately fit.

Instead, direct estimates of poverty rates from survey data y_i^d are used as the outcome variable. These estimates are nevertheless subject to sampling error because they are based on the area-specific sample data. The second part of the model thus assumes that the direct estimates are centered around the true poverty rates as follows:

$$y_i^d = y_i^a + \omega_i \quad (7)$$

where ω_i represents measurement or sampling error and is assumed to have heteroskedastic variance $\sigma_{\omega,i}^2$ because sample sizes across areas are often unequal. These variances are assumed to be known, although in

practice they are estimated using the survey data. Given the two parts of the model, the survey data can then be used to obtain parameter estimates.

The best linear unbiased predictor (BLUP) is obtained by predicting poverty rates through the model $x_i\hat{\beta} + \hat{\varepsilon}_i$ where $\hat{\beta}$ and $\hat{\varepsilon}_i$ represent the estimated coefficients and area-level effects, respectively. More specifically, the estimated area effect is $\hat{\varepsilon}_i = \psi_i (y_i^d - x_i\hat{\beta})$ where $\psi_i = \hat{\sigma}_\varepsilon^2 / (\hat{\sigma}_\varepsilon^2 + \hat{\sigma}_{\omega,i}^2)$. In essence, the final estimates for areas contained in the survey are the weighted average between the direct estimate y_i^d and the synthetic estimator $x_i\hat{\beta}$ where the weights, ψ_i and $(1 - \psi_i)$, are determined by the quality of the model and the sample sizes for each area. For the non-sampled entities, the area-level model simply uses the synthetic estimator to produce predictions. A critical feature of the area-level model is that it is a direct competitor to the machine-learning approach because both methods are intended for use in off-census years and do not rely on household-level data.

In summary, we use both the unit-level and area-level models as benchmarks in our assessment of the modern approach to poverty mapping. While the “gold standard” unit-level model provides an absolute performance benchmark by approximating the upper performance bound, the area-level model provides a relative performance benchmark as the leading competitor to the modern approach. In our implementations of these models, we follow [Corral et al. \(2022\)](#) and remove any eligible covariate that is weak or extraneous. For example, when fitting the area-level model on a given sample, we sequentially remove insignificant covariates from the set of eligible variables, beginning with the one with the highest p -value. We then address collinearity by removing any variable with a variance inflation factor above five.¹³ All covariates enter into this procedure linearly because identifying transformations (e.g., quadratic or cubic terms) would substantially complicate our sampling experiments.

3. Model validation

The coefficient of determination or R^2 is the most commonly used metric for validating poverty maps generated using machine-learning methods.¹⁴ In this section, we first critically assess this validation approach by discussing three ways in which the R^2 can be misleading in terms of model assessment. We then motivate our simulation experiments by proposing an alternative approach that remedies many of the limitations of the standard procedure. Specifically, we propose (1) assessing model performance relative to true poverty measures rather than direct estimates, and (2) using mean squared error and targeting performance as the principal validation metrics. Our objective is to present a validation procedure that is appropriate for our purposes rather than a general strategy that is feasible for all poverty-mapping exercises.¹⁵

Regarding our assessment of the standard procedure, let y_i^a denote the actual or true poverty indicator for the i th geographic entity. Further, let y_i^d and y_i^p denote the direct estimate and model-based prediction of the poverty indicator, respectively. The standard method for assessing the predictive performance of model-based poverty estimates calculates the R^2 from a regression like the following:

$$y_i^d = \alpha + \delta y_i^p + \zeta_i \quad (8)$$

¹³ See Section 3.2 in [Corral et al. \(2022\)](#) for additional discussion. Variable selection for the unit-level model is slightly more complicated, as the above procedure is preceded by a lasso regression. See Section 6.2 in [Corral et al. \(2022\)](#) for details.

¹⁴ See [Jean et al. \(2016\)](#), [Pokhriyal and Jacques \(2017\)](#), [Steele et al. \(2017\)](#), [Lee and Braithwaite \(2020\)](#), [Yeh et al. \(2020\)](#), or [Chi et al. \(2022\)](#).

¹⁵ For example, our validation procedure requires a dataset that is sufficiently large for repeated sampling and such data is unlikely to be available in many contexts.

¹⁰ See [Corral et al. \(2021b\)](#) for a more detailed description.

¹¹ As opposed to empirical best estimates, census empirical best estimates do not include survey observations when calculating area-level poverty rates. [Corral et al. \(2021b\)](#) show that when the sample size is small relative to the population, the difference between the two methods is negligible. [Corral et al. \(2021b\)](#) also show that the census empirical best method performs quite well in simulations, particularly in terms of tracking the variance of the true poverty rates. This contrasts with ELL-type methods, which have a tendency to underestimate the variance. For further discussion of the census empirical best method, also see [Molina \(2019\)](#).

¹² While we focus on the original formulation of the model, note that several extensions have been proposed, particularly to accommodate more complicated error structures (e.g., spatial correlation). See Chapter 3 of [Corral et al. \(2022\)](#) for discussion.

where α and δ are the regression coefficients and ζ represents the error term.¹⁶ The R^2 for the above regression can be written as follows:

$$R_d^2 = \frac{\sigma_d^2 - \sigma_\zeta^2}{\sigma_d^2} \quad (9)$$

where σ_d^2 is the variance of the direct estimate and σ_ζ^2 is the variance of ζ . R_d^2 conveys information about how much of the variance of the direct estimate is explained by the model-based prediction.

While unbiased, direct estimates can be imprecise estimates of true poverty indicators (Corral et al., 2021b). Ideally, one would assess the performance of model-based predictions on the basis of the true poverty indicators themselves. That is, model-based predictions are more appropriately assessed using the R^2 from the following regression:

$$y_i^a = \alpha + \delta y_i^p + v_i \quad (10)$$

where v represents the error term. The R^2 for this regression can be written as

$$R_a^2 = \frac{\sigma_a^2 - \sigma_v^2}{\sigma_a^2} \quad (11)$$

where σ_a^2 is the variance of the true poverty indicator and σ_v^2 is the variance of v . We are first interested in characterizing the bias of R_d^2 when it is used to approximate R_a^2 .

Let $y_i^d = y_i^a + \omega_i$ where ω is a mean-zero error term with variance σ_ω^2 . That is, we view this as a classical measurement error problem where the direct estimate of the poverty indicator is a noisy estimate of the true poverty indicator.¹⁷ Note that the preceding equation implies that $\zeta_i = v_i + \omega_i$ and $\sigma_\zeta^2 = \sigma_v^2 + \sigma_\omega^2$. Following Majeske et al. (2010), we then define the bias as:

$$B = R_a^2 - R_d^2. \quad (12)$$

Substituting Eqs. (9) and (11) into our expression for the bias, we have:

$$B = \frac{\sigma_a^2 - \sigma_v^2}{\sigma_a^2} - \frac{\sigma_d^2 - \sigma_v^2 - \sigma_\omega^2}{\sigma_d^2} = \frac{\sigma_\omega^2(\sigma_a^2 - \sigma_v^2)}{\sigma_d^2 \sigma_a^2} = \frac{\sigma_\omega^2}{\sigma_d^2} R_a^2. \quad (13)$$

Finally, we can substitute Eq. (13) into Eq. (12) and rearrange, which gives

$$R_d^2 = R_a^2 \left(\frac{\sigma_d^2 - \sigma_\omega^2}{\sigma_d^2} \right) \quad (14)$$

and establishes a basic relationship between R_d^2 and R_a^2 . See Majeske et al. (2010) for additional mathematical details.

In Eq. (14) above, we see that $R_d^2 = R_a^2$ when $\sigma_\omega^2 = 0$ or, equivalently, when $y_i^d = y_i^a$ for all i . In the presence of measurement error or when $\sigma_\omega^2 > 0$, we then see that R_d^2 will be biased downward and the magnitude of the bias will be determined by the size of $(\sigma_d^2 - \sigma_\omega^2)/\sigma_d^2$. Validation exercises based on R_d^2 will thus tend to mischaracterize the performance of poverty maps, with the resulting estimates of R^2 being biased downward. One implication of this bias is that R_d^2 is context-dependent and will generally be influenced by the design of the survey data (e.g., larger sample sizes will tend to reduce measurement error, all else equal). In particular, a poorly performing model applied to a low bias setting may yield an estimate of R_d^2 that is higher than an estimate of R_d^2 from a strong model applied to a high bias setting. The presence of bias in the measures of R^2 can thus reverse the rank ordering of models when comparisons are made across contexts.¹⁸

¹⁶ Note that any regression of this form can only be estimated for the sampled regions with direct estimates.

¹⁷ The assumption of classical measurement error may be violated in practice. For example, if poorer households are less likely to be sampled, the measurement error will not be independent of the true poverty rates.

¹⁸ See Yeh et al. (2020), Chi et al. (2022), or Aiken et al. (2022) for this type of cross-context comparison of R^2 values.

There is a second form of bias in R^2 that can conversely lead to overoptimistic conclusions regarding model validity. To illustrate this, note that R_a^2 in Eq. (11) is mathematically equivalent to the squared (Pearson) correlation coefficient between y_i^a and y_i^p . A well-known property of the correlation coefficient is that it is invariant to any positive affine transformation of its arguments, implying that the squared correlation coefficient is invariant to any affine transformation (Gujarati, 2003).¹⁹ We can thus write

$$R_a^2(y_i^a, y_i^p) = R_a^2(y_i^a, q_1 + q_2 y_i^p) \quad (15)$$

where q_1 and q_2 are some arbitrary constants. The implication of the above is that the R^2 is insensitive to systematic bias in the model-based poverty estimates. Consider some unbiased poverty estimate y_i^u , meaning that $E(y_i^u - y_i^a) = 0$. Now consider another estimate y_i^b that is identical to y_i^u up to some locational shift, implying that $y_i^b = y_i^u + q$ and $E(y_i^b - y_i^a) = q$. While y_i^b is biased (i.e., it will systematically over- or under-estimate poverty), it will misleadingly achieve an R^2 that is identical to y_i^u .

A third issue related to R^2 as a performance metric is that it is affected by the variance of the true poverty measures. To see this, consider the following re-expression of R_a^2 :

$$R_a^2 = 1 - \frac{\frac{1}{N} \sum_i (y_i^a - \alpha - \delta y_i^p)^2}{\frac{1}{N} \sum_i (y_i^a - \bar{y}^a)^2} = 1 - \frac{\text{MSE}(y_i^a, \alpha + \delta y_i^p)}{\text{Var}(y_i^a)} \quad (16)$$

where N is the number of geographic entities. The above shows that R^2 can be viewed as a normalized version of the mean squared error with the variance of the true poverty indicator used as the normalization factor. This approach to normalization produces an additional source of context-dependence in R^2 . Consider two models that perform equally well in that they achieve the same mean squared error. However, assume that one model is applied to a context where the variance of y_i^a is large and the other is applied to a low-variance context. The above implies that the model applied to the high-variance context will achieve a higher R^2 , even though the performance of the two models is arguably identical.

The limitations of the standard procedure stem both from the R^2 metric itself and from applying the R^2 to direct poverty estimates rather than true poverty indicators. Accordingly, we address these limitations by making two modifications to the usual procedure. To address the issue related to error in the reference measure, we assess our poverty-mapping methods with data that allows us to use true poverty indicators as the reference measure (see the next section for discussion). We then address the remaining limitations by focusing on validation metrics that are sensitive to systematic biases in the poverty estimates and avoid the normalization issue. To be clear, one could alternatively attempt to correct the R^2 for its biases, but any adjustments would require approximations that could lessen the credibility of our validation exercises.²⁰

Regarding our validation metrics, the mean squared error is closely related to the R^2 , but sidesteps its most critical limitations. The mean squared error is defined as follows:

$$\text{MSE}(y_i^a, y_i^p) = \frac{1}{N} \sum_i (y_i^a - y_i^p)^2 \quad (17)$$

and we immediately see that it provides a direct understanding of the expected squared error of the estimates and thus avoids the normalization issue. It is also sensitive to systematic biases in the predictions. Reconsidering the biased and unbiased poverty estimates described

¹⁹ More specifically, the correlation coefficient is invariant to any positive affine transformation because it changes signs when one of its arguments is scaled by some negative constant. The squared correlation coefficient is then invariant to any affine transformation.

²⁰ For example, correcting the bias associated with the measurement error issue would require estimating the variance of the measurement error.

above (i.e., y_i^u and y_i^b), we can write $\text{MSE}(y_i^b, y_i^a) = \text{MSE}(y_i^u + q, y_i^a) = \text{MSE}(y_i^u, y_i^a) + q^2$. Systematic biases in the estimates will then be appropriately penalized by the mean squared error. Another advantage is that with multiple estimates for each entity, one can calculate entity-specific mean squared errors, which can be decomposed into parts associated with the bias and variance of the estimates. Where y_{ih}^p denotes estimate $h = 1, 2, \dots, H$ for the i th entity, we have

$$\text{MSE}_i = \frac{1}{H} \sum_h (y_i^a - y_{ih}^p)^2 = \frac{1}{H} \sum_h (y_{ih}^p - \bar{y}_i^p)^2 + (\bar{y}_i^p - y_i^a)^2 \quad (18)$$

where the first term on the right-hand side of the above captures the variance of the estimates for entity i and the second term captures the (squared) bias.²¹

While calculating the error of the poverty estimates is useful for understanding the accuracy of poverty maps, the primary reason for obtaining granular poverty estimates is to improve the targeting of resources. As such, we complement the mean squared error with an alternative performance metric that focuses on how different methodological choices affect poverty alleviation through influencing the efficiency of targeting. Similar to [Elbers et al. \(2007\)](#), we simulate a hypothetical poverty-alleviation program that distributes resources to households according to a given poverty map and then calculates the amount of poverty-reduction associated with the map.²² Though using targeting performance as a validation metric has direct policy relevance, note that it can be insensitive to systematic biases in the poverty estimates (e.g., rank-based targeting schemes do not depend on the absolute level of the poverty estimates).

The procedure we use in our simulations is as follows:

1. Using true household-level incomes, calculate the poverty headcount and poverty gap for the country as a whole for some chosen poverty line. Using the poverty gap measure, determine the budget required to completely eradicate poverty. Calculate an individual-level transfer amount by dividing the total budget by the number of people living in poverty.
2. Using a given poverty map, order geographic areas according to their estimated poverty headcounts, from most to least poor. For the poorest area, augment the true per capita income of all households by an amount equal to the transfer (i.e., each household receives an amount equal to the transfer times the number of household members). Continue this distribution procedure for the second poorest area, the third poorest area, and so on until there are not enough resources to cover all households in a given area. For the marginal area, distribute the remaining budget equally among individuals in that area.
3. Using the post-transfer household incomes, calculate an updated poverty headcount to assess the effects of the distribution scheme on poverty alleviation.
4. Repeat steps 2 and 3 for each of the poverty maps generated by the alternative methods.

The benchmark poverty alleviation from this scheme corresponds to the results of applying the above to the true poverty rankings.

4. Data

Carried out by the Mexican National Institute of Statistics and Geography (INEGI) in 2015, the Mexican Intercensal Survey is a unique dataset that allows us to construct a ground truth for our validation exercises. The sample consists of 5.9 million households and is representative at the national, state (32 states), and municipality level (2457

municipalities). It is also representative for localities with populations of 50,000 or more inhabitants. The survey questionnaire gathered information on household income, geographic location, household demographics, and dwelling characteristics, among other things.²³ The size of the dataset is especially important, as it allows us to conduct simulation experiments with repeated samples that are sufficiently large and diverse. We use these samples to estimate municipality-level poverty maps with alternative methods and then assess the maps using the full dataset as the ground truth.

Prior to sampling from the intercensus, we make three modifications to the data. First, as a large number of households reported earning no income, we randomly remove 90 percent of these households.²⁴ Second, to ensure that all municipalities are sufficiently large, municipalities with less than 500 households are removed. Finally, to ensure that all primary sampling units (PSUs) are also sufficiently large for sampling purposes, we combine several neighboring PSUs such that all include at least 300 households. Our final census dataset consists of 3.9 million households in 1,865 municipalities and 16,297 PSUs. The resulting census is used to draw 500 survey samples that serve as the basis for our simulation experiments ([Tzavidis et al., 2018](#)). In what follows, we describe our approach to constructing these samples.

Our sampling procedure is intended to reflect standard design elements used in many face-to-face household surveys conducted in the developing world ([Grosh and Muñoz, 1996](#)). Mexico's 32 states comprise the intended domains of the sample and the indicator of interest is household income per capita. For each sample, we seek to achieve a relative standard error (RSE) of 7.5 percent for average household income in each state.²⁵ We use a two-stage clustered design wherein PSUs or clusters are selected within each domain and then a sample of 10 households is selected within each cluster. Our decision to select 10 households within each cluster is consistent with the design of other large-scale household surveys conducted in developing countries ([Grosh and Muñoz, 1996](#)). Under simple random sampling, the minimum sample size required for each state is as follows:

$$n_s = \left(\frac{\sigma_s}{\bar{w}_s \times \text{RSE}} \right)^2 \quad (19)$$

where $\bar{w}_s$ and σ_s represent average household income per capita and the standard deviation of per capita income for state s , respectively.

The minimum sample size under simple random sampling must be adjusted to account for the clustered design. The design effect due to clustering is accounted for by estimating the intra-cluster correlation of per capita income within each state. The correlation estimates can then be used to adjust the above sample sizes for the clustered design as follows:

$$n_s \times \left[1 + \rho_s (n_{sc} - 1) \right] \quad (20)$$

where n_{sc} denotes the number of households selected in cluster c within state s (10 in this case) and ρ_s is the intra-cluster correlation of per capita income. Given the above, we can calculate the number of clusters needed to achieve the minimum sample size for each state and then multiply this number by 10 to find the final sample size. Taking these sample size requirements as given, each of our samples is then drawn in accordance with the two-stage design.

Specifically, the clusters within each state are selected with probability proportional to size (without replacement) and the households within each cluster are selected via simple random sampling. According

²¹ See [Hastie et al. \(2009\)](#) for derivation and detailed discussion of the bias-variance decomposition.

²² See [Yeh et al. \(2020\)](#), [Chi et al. \(2022\)](#), and [Aiken et al. \(2022\)](#) for examples of other studies that have conducted targeting simulations.

²³ Income is defined as money received from work performed during the course of the reference period by individuals aged 12 or older.

²⁴ This is done to ensure some missing values are present in the data used, but not as many as in the original data.

²⁵ The desired precision of 7.5 percent is somewhat arbitrary, but corresponds to precision targets in similar surveys and yields samples of reasonable sizes.

Table 1
Select descriptive statistics for the Mexican Intercensal Survey.

Variable	Mean	Std. Dev.	Min.	Max.	Obs.
Municipality poverty rate (full intercensus)	0.30	0.19	0.02	0.96	1,865
PSU poverty rate (full intercensus)	0.23	0.20	0.00	1.00	16,297
Observations per sample	23,540	0.00	23,540	23,540	500
Municipalities per sample	992.04	11.95	951	1,020	500
PSU-to-municipality ratio per sample	2.37	0.03	2.31	2.48	500

to this design, the inclusion probability for a given household is then approximately:

$$\frac{\tilde{n}_{sc} N_s}{\tilde{n}_s} \times \frac{n_{sc}}{\tilde{n}_{sc}} \quad (21)$$

where $\tilde{n}_{sc}$ is the total number of households in cluster c within state s , N_s is the number of clusters selected in state s , and $\tilde{n}_s$ is the total number of households in state s .²⁶ Table 1 presents select descriptive statistics related to the full intercensus and the 500 samples. We set the poverty line at the 25th percentile of the true income distribution and the first two rows summarize the associated poverty rates at the municipality and PSU levels for the full dataset. The remaining rows summarize our samples and show that we draw 23,540 households per sample (2354 PSUs), with an average number of municipalities per sample of roughly 992. The samples thus have an average of 2.37 PSUs per municipality.

In addition to the intercensus, INEGI has also released geo-referenced data that is similar to the remotely-sensed covariates often used in machine-learning applications. These geographic variables were calculated by INEGI for the year 2015 (the same year as the intercensus) and the municipality is the lowest administrative level for which the covariates are available.²⁷ The geo-referenced data includes 105 indicators. Specifically, the dataset includes information from 21 different dimensions and each dimension is associated with five summary measures.²⁸ For example, we summarize the atmospherically resistant vegetation index for each municipality using the minimum, maximum, mean, sum, and standard deviation of the index. The same procedure is applied to all dimensions. The geo-referenced data has one major limitation, which is that it does not include information on night-time lights, which is a commonly-used covariate in machine-learning applications.²⁹

²⁶ The equation used to calculate inclusion probabilities assumes sampling with replacement, but is used here as an approximation of inclusion probabilities under proportional selection without replacement. This should provide a reasonable approximation in this case since there are a relatively large number of clusters present in the frame. The design weight for each household is simply the inverse of the inclusion probability. In a typical survey, the design weights would be further adjusted for nonresponse and calibrated to known population characteristics. However, since the sampling is only a simulation exercise, there is no nonresponse and thus no nonresponse adjustment is required. Calibration or post-stratification could be performed, but was not implemented to simplify the process.

²⁷ Technically speaking, the intercensus does not include identifiers that provide information on geographic location below the municipality level. We are thus only able to merge the intercensus and geo-referenced data at the municipality level.

²⁸ The 21 dimensions are as follows: enhanced vegetation index, normalized difference vegetation index, normalized difference built-up index, built-up index, simple ratio, atmospherically resistant vegetation index, urban index, index of biological integrity, normalized difference water index, modified normalized difference water index, new built-up index, band ratio for built-up area, normalized built-up area index, built-up area extraction index, normalized difference snow index, visible atmospherically resistant index, soil-adjusted vegetation index, optimized soil-adjusted vegetation index, normalized difference moisture index, digital altitude model, and digital slope model.

²⁹ For examples, see Jean et al. (2016), Lee and Braithwaite (2020), or Chi et al. (2022).

We thus augment the INEGI geo-referenced data with night-time lights data produced by the Earth Observation Group (EOG). EOG uses the nightly day/night band (DNB) low light imaging data gathered by the NASA/NOAA Visible Infrared Imaging Radiometer Suite (VIIRS) (Elvidge et al., 2021). To match the other geo-referenced covariates, we use the night-time lights data from 2015, which is available at 15 arc-second resolution (roughly 500 meters at the equator) and then aggregated to the municipality level. Also like the other geo-referenced data, we summarize night-time lights within each municipality using the minimum, maximum, mean, sum, and standard deviation. With these additions, we then have two alternative sets of covariates that we use in our simulations: the remotely-sensed data (including night-time lights) and the standard set of sociodemographic covariates from the intercensus.

5. Results

Recall that our primary objective is to assess the performance of the modern approach to poverty mapping by using the Mexican Intercensal Survey as a ground truth. To this end, panel (a) in Fig. 1 presents the mean squared error (MSE) for our closest approximation to the standard implementation of the modern approach: gradient boosting with remotely-sensed covariates (covariates labeled “GIS”). We benchmark the performance of this implementation against two traditional mapping methods, namely the unit-level and area-level models. To approximate how these traditional methods are commonly used, each model is implemented with census-based covariates, albeit at different levels of analysis (covariates labeled “CEN”). Importantly, as our initial implementations are intended to reflect current practice, the modern and traditional approaches differ both in terms of their statistical method and the covariates used. The remainder of this section will examine how these differences contribute to performance disparities between the two approaches.

Panel (a) of Fig. 1 shows that gradient boosting with remotely-sensed covariates tends to have the highest MSE of the models considered. The associated boxplot summarizes the distribution of MSE values across all municipalities (i.e., the results are based on Eq. (18)) and indicates that the gradient-boosting model achieves a median MSE of 0.007. The corresponding root MSE is approximately 0.08, meaning that the median municipality’s error is roughly eight percentage points. By contrast, the “gold standard” unit-level achieves a median MSE of 0.003, which is roughly 40 percent that of the gradient-boosting model. The area-level model performs similarly to the unit-level model, with a median MSE of 0.004. The fact that the area-level model outperforms the gradient-boosting model is particularly notable, as both of these models were developed for use in off-census years and are thus direct competitors.

Panel (b) of Fig. 1 considers the targeting performance of the alternative models and shows a similar result. For these simulations, we have set the poverty line at the 25th percentile of the (pre-transfer) per capita income distribution, which means that the baseline poverty rate is 25 percent. The poverty rate falls to 19.9 percent when transfers are delivered to municipalities according to their true poverty rate. Across 500 simulations, we find that gradient boosting with remotely-sensed covariates achieves a median poverty rate of 20.29, which is close to the benchmark rate of 19.9 percent. The two traditional methods nevertheless outperform the gradient-boosting model. The unit-level

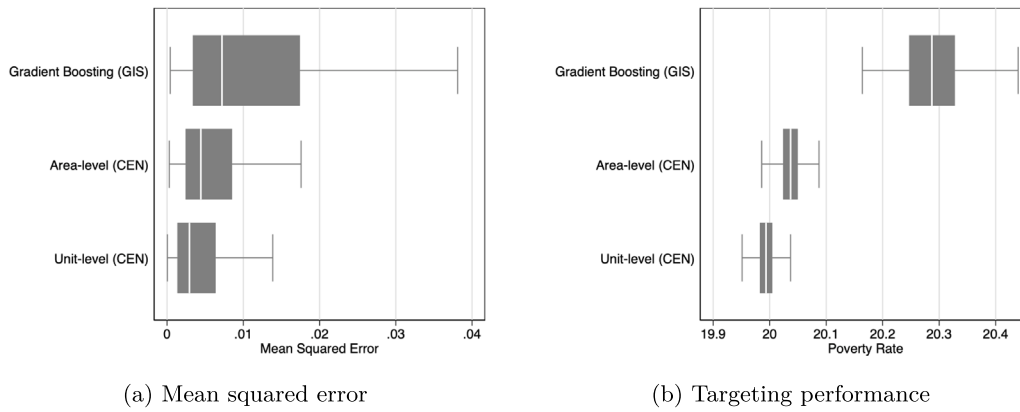


Fig. 1. Mean squared error and targeting performance for select implementations.

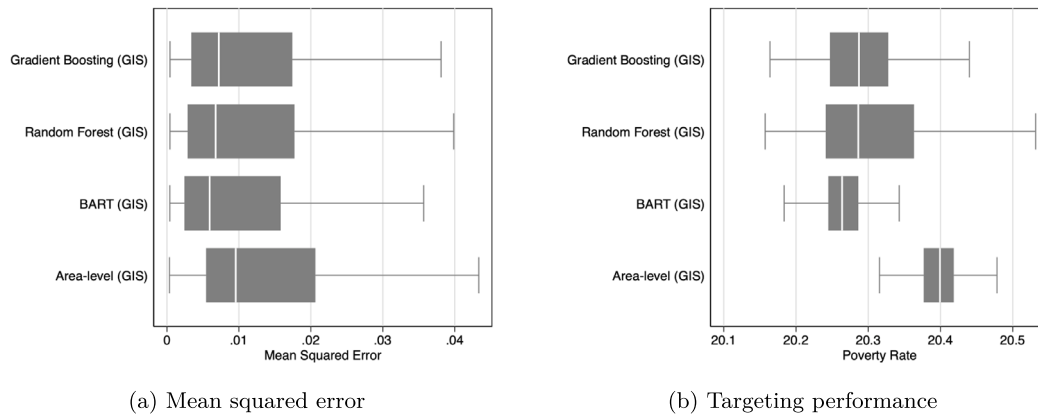


Fig. 2. Mean squared error and targeting performance for alternative statistical methods .

model achieves a median poverty rate of 20 percent whereas the area-level model achieves a median rate of 20.04. The traditional approach thus outperforms the modern approach by about 0.3 percentage points, which in a country the size of Mexico corresponds to about 360,000 additional people escaping poverty.³⁰

As mentioned, there are two primary differences between the traditional and modern approaches that could explain these performance disparities: the statistical method and the covariates used. We first examine whether using an alternative statistical method could improve upon the performance of our initial implementation of the modern approach. Fig. 2 presents MSE and targeting results for four alternative statistical methods where each uses the same, remotely-sensed covariates in order to isolate the influence of the method. In addition to gradient boosting, we consider two leading non-parametric competitors (i.e., random forest and Bayesian additive regression trees) and the leading parametric competitor (i.e., the area-level model).³¹ We find that the alternative methods do not produce any substantive performance gains. In particular, the two alternative non-parametric methods perform similarly to gradient boosting whereas the parametric, area-level model tends to perform slightly worse when using the same covariates.³²

By contrast, Fig. 3 shows that implementing gradient boosting with census-based covariates leads to notable performance gains. The first two boxplots in panels (a) and (b) present the results for gradient boosting with remotely-sensed and census-based covariates, respectively. As a benchmark, we plot similar results for the area-level model. Recall that when using remotely-sensed covariates, gradient boosting achieves a median MSE of 0.007 and a median poverty rate of 20.29. With census-based covariates, the median MSE falls to 0.003 and the median poverty rate falls to 20.02. Importantly, gradient boosting with census-based covariates slightly outperforms the area-level model, which achieves a median MSE of 0.004 and a median poverty rate of 20.04 when using the same covariates. The under-performance of the modern approach is thus largely driven by the remotely-sensed covariates.³³

Another way to see that the census-based covariates have greater predictive capacity than the remotely-sensed covariates is by looking at feature importance. One popular measure of feature importance summarizes the gain (i.e., reduction of the loss function) associated with any given feature. That is, for a given run of the model, feature importance captures the average gain of a given feature, where the gains are averaged across all trees in the model.³⁴ Fig. 4 plots the 20

³⁰ Mexico's population was about 120 million in 2015, which is the year that the Mexican Intercensal Survey was conducted.

³¹ In terms of non-parametric models, both random forests and Bayesian additive regression trees are quite popular and have been shown to perform comparably to, and sometimes better than, gradient boosting. For example, see Chipman et al. (2010).

³² Bayesian additive regression trees perform slightly better than gradient boosting, but the performance gains are not large. For example, in terms of

targeting performance, gradient boosting achieves a median poverty rate of 20.29 whereas the Bayesian method achieves a median rate of 20.26.

³³ This result is similar to van der Weide et al. (2022) who find that remotely-sensed covariates lose significance once census-based covariates are added to their model.

³⁴ In addition, the average gains for a given run of the model are normalized by dividing each by the sum of all average gains for that run.

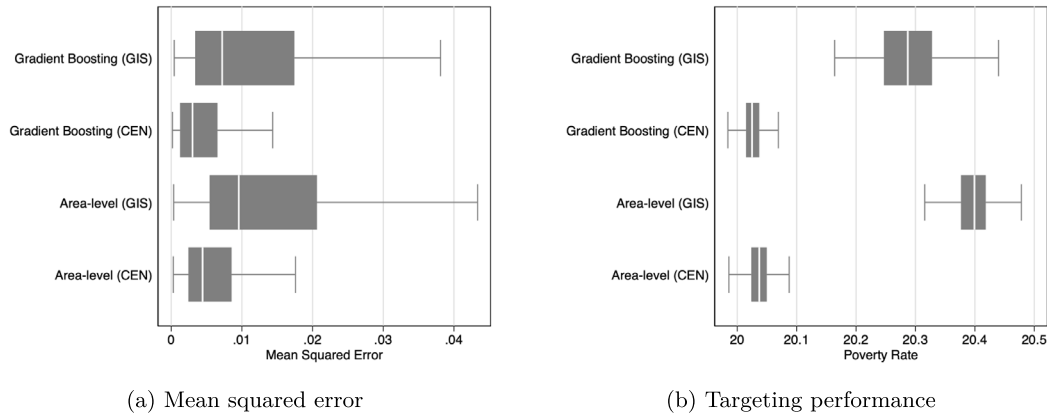


Fig. 3. Mean squared error and targeting performance for alternative covariates .

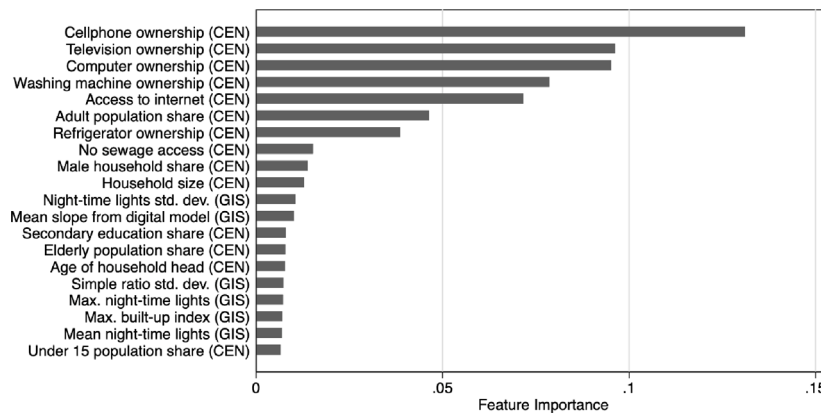


Fig. 4. Feature importance.

covariates with the highest feature importance after running gradient boosting with both census-based (“CEN”) and remotely-sensed (“GIS”) covariates on each sample and averaging each covariate’s importance across the 500 samples. We find that all of the top 10 covariates are census-based, with only six remotely-sensed covariates entering the top 20. Note also that the feature importances outside the top 10 are relatively small in magnitude.

The above results suggest that gradient boosting can approximate the performance of the unit-level model and outperform the area-level model when using the same covariates. The aggregated nature of our performance comparisons thus far, however, mask some important differences between the methods. To illustrate these differences, we first use the variance-bias decomposition presented in Eq. (18) and plot the two components independently. Panel (a) of Fig. 5 presents box-plots summarizing the distribution of the variance component across municipalities for select models. We see that gradient boosting is associated with lower variance estimates than the area-level model for any given set of covariates. For example, the median variance for gradient boosting with remotely-sensed covariates is 0.002 whereas the median variance for the area-level model with the same covariates is 0.005. This variance reduction is an important advantage of gradient boosting and the reason for its use of regularization.

Panel (b) presents the bias component of the decomposition for the same set of models and shows that all the models depicted are approximately unbiased at the median.³⁵ The models using remotely-sensed

covariates nevertheless exhibit wide variation in the bias estimates, implying that the estimated poverty rates for some municipalities are severely biased. For example, the minimum and maximum bias estimates for gradient boosting with remotely-sensed covariates are -45 and 40 percentage points, respectively. The area-level model exhibits similar variation in the bias estimates when using remotely-sensed covariates, indicating that weak predictors can produce considerable biases for both methods, at least for some municipalities. While gradient boosting and the area-level model appear to be similarly biased for any given covariate set, a closer look at the source of these biases reveals an important shortcoming of gradient boosting.

To this end, panel (a) of Fig. 6 plots the municipality-level bias estimates from select models on the true municipality poverty rates. For each model, we fit a lowess curve to the data to show the (weighted) average of the bias estimates at different true poverty rates. The figure shows the source of the aforementioned bias variability: both models with remotely-sensed covariates severely overestimate poverty in the least poor municipalities and underestimate poverty in the most poor municipalities. Stated differently, the models with remotely-sensed covariates understate the spatial distribution of poverty. This result occurs because the predictions collapse toward the mean in the absence of informative covariates, which produces predictable biases in the tails of the distribution. Interestingly, for both covariate types, the area-level model is less biased for the poorest municipalities.

Panel (b) in Fig. 6 shows that an important reason for the area-level model’s reduced bias is that its in-sample predictions are a weighted average of the model-based estimates and the (unbiased) direct esti-

³⁵ The variance-bias decomposition shows that the MSE is the sum of the variance and the squared bias of the estimates. To illustrate both underestimates and overestimates, we do not square the bias estimates.

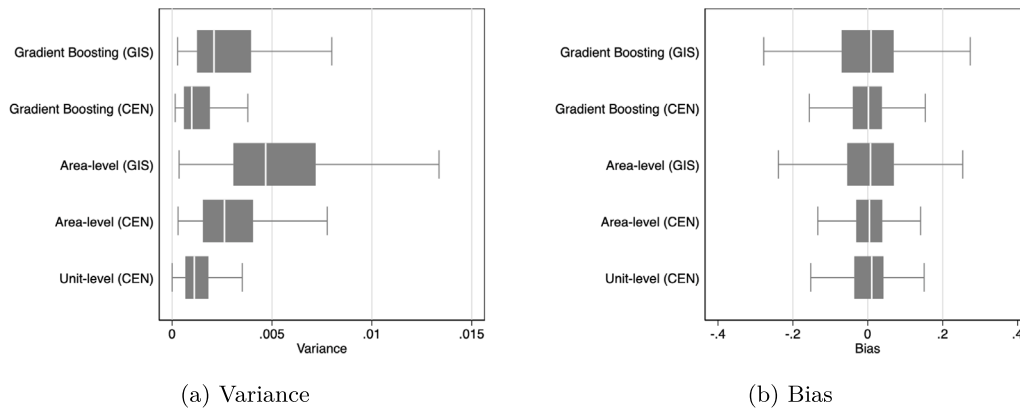


Fig. 5. Variance-bias decomposition for select models .

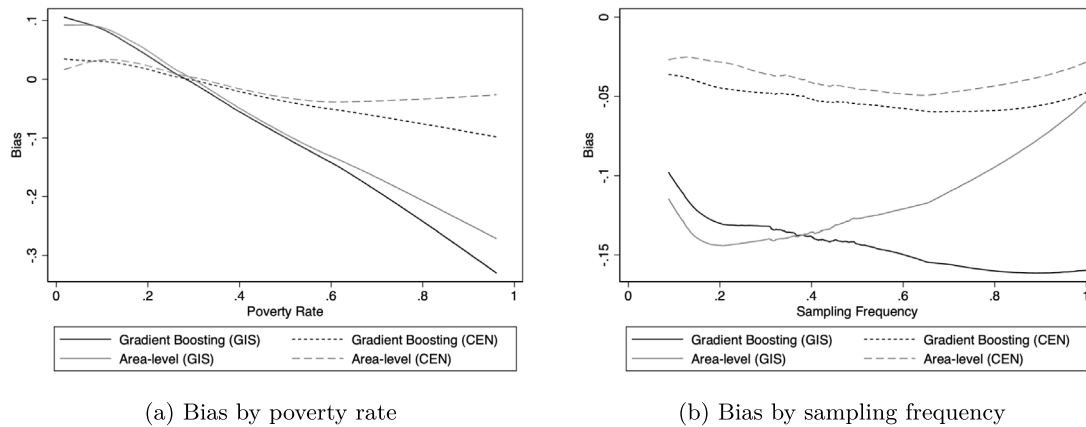


Fig. 6. Bias by poverty rate and sampling frequency.

mates.³⁶ For each model depicted, we focus on the 20 percent of municipalities with the highest true poverty rate, plot the municipality-level bias estimates on the frequency with which a municipality was sampled, and then fit a loess curve to the data. The sampling frequencies capture the proportion of samples for which a municipality received an in-sample prediction, thus benefitting from the area-level model's averaging procedure. As expected, we see that the area-level model is less biased for municipalities with higher sampling frequencies. The performance disparity is especially pronounced among the models with remotely-sensed covariates, as the area-level model is roughly 10 percentage points less biased at the highest sampling frequencies.

These latter results highlight an important methodological difference between the modern and traditional approaches. In particular, the machine-learning methods used for poverty mapping were developed for out-of-sample prediction and ignore the direct estimates after fitting the model. The traditional methods, by contrast, were developed specifically for small area estimates of poverty and recognize that the estimation is partly an in-sample prediction problem. The traditional methods thus use the direct estimates after fitting the model and weight them more heavily when the covariates are less informative (see Section 2 for discussion of the BLUP method). Interestingly, this highlights another concern with using direct estimates to assess model performance: traditional methods are designed to be less predictive of direct estimates when the model is more informative. The main point here is, nevertheless, that our results are in line with the expectation

that traditional methods will exhibit less in-sample bias, particularly in the presence of weak covariates.

6. Conclusion

Machine-learning approaches to poverty mapping have become increasingly popular in recent years. In contrast to traditional mapping procedures, modern machine-learning methods rely on non-parametric statistical techniques applied to remotely-sensed data, such as satellite imagery or call detail records. In addition, machine-learning approaches routinely conduct model assessment using the R^2 , which is often calculated by regressing direct, survey-based poverty estimates on model predictions. While direct estimates are unbiased, they can be imprecise measures of true poverty rates and, as a result, it is unclear whether this approach to assessment is informative of model performance. In this paper, we thus conducted a more rigorous assessment of the machine-learning approach to poverty mapping by using the Mexican Intercensal Survey for several design-based simulation experiments.

To motivate our experiments, we outlined several ways that the R^2 can be a misleading measure of model performance: it is biased downward when direct estimates are used as the reference measure of model performance, it is insensitive to differences in location and scale between the reference measure and the model predictions, and it is normalized by the variance of the reference measure. The limitations of the standard validation procedure are due both to the R^2 metric itself and from applying the R^2 to direct poverty estimates rather than true poverty measures. Our experiments address these limitations in two ways: (1) we assess model performance using data that allows us to use true poverty indicators as the reference measures, and (2) we

³⁶ See Section 2 for details. By contrast, gradient boosting relies exclusively on the model-based estimates and only incorporates the direct estimates when fitting the model.

emphasize validation metrics that are sensitive to systematic biases in the poverty estimates and avoid the normalization issue.

Our results showed that the modern approach to poverty mapping under-performs relative to the traditional approach both in terms of mean squared error and poverty reduction. We found that this under-performance is driven by the modern approach's reliance on remotely-sensed covariates. When using the same covariates, machine learning performs comparably to the unit-level model and slightly outperforms the area-level model. We then showed that the performance advantage of machine learning relative to the area-level model is due to its variance-reduction properties, namely its use of regularization. The cost of such variance reduction, however, is that machine learning exhibits more bias in the tails of the poverty distribution than the area-level model. This bias-variance tradeoff is especially stark in the context of remotely-sensed covariates, which suggests that the quality of the available data is an important consideration when choosing between parametric and non-parametric mapping methods.

Our analysis has several practical implications for poverty mapping. For one, producers and users of poverty maps should exercise caution when using R^2 as a model assessment tool, especially when it is calculated using direct poverty estimates as the reference measure. Poverty maps should ideally be validated using a credible ground truth and alternative validation metrics, such as the mean squared error and targeting performance. As the survey data used in our experiments is uniquely large, researchers may need to consider alternative design-based validation exercises. Tzavidis et al. (2018) recommend using available census data to conduct sampling experiments based on a variable that is correlated with the variable of interest. Corral et al. (2021a) alternatively recommend using census data to conduct "hybrid" sampling experiments using a synthetic dependent variable.³⁷

Our results also highlight that covariate quality has important implications for the practice of poverty mapping. Most obviously, the limited predictive capacity of remotely-sensed covariates suggests that researchers should incorporate census-based covariates whenever possible. Perhaps more interestingly, our results suggest that the quality of the available covariates has implications for the choice of statistical model, particularly in off-census years. With high-quality covariates, there are only minor performance differences between the non-parametric and parametric approaches to estimating poverty maps. Machine learning is a reasonable choice in this case, as it performs slightly better overall with only minimal increases in bias. With lower-quality covariates, however, the modest overall advantage of machine learning is coupled with substantial increases in bias for some areas. Machine learning should be used with caution in this case, especially when poverty maps are needed to understand absolute poverty levels.

Stated differently, there are important tradeoffs to consider when choosing between modern and traditional poverty-mapping methods, particularly in off-census years. While both approaches require survey-based welfare measures, producers of poverty maps must weigh the costs of gathering different types of covariate data against their relative performance benefits. For example, we found that the traditional approach only slightly outperforms the modern approach in terms of targeting performance. The modern approach could potentially be a reasonable choice if there are substantial costs associated with accessing census data. Our targeting results nevertheless only depend on each method's rank-ordering of areas and do not consider the levels of poverty estimated by each method. In cases where poverty levels are relevant (e.g., determining program quotas), producers of poverty maps must additionally navigate the bias-variance tradeoff.

We have identified several limitations of the modern approach to poverty mapping, but our analysis has its own drawbacks. We have focused on measuring income poverty and have not considered

the performance of the modern approach when used in the context of other welfare measures (e.g., wealth).³⁸ We have also only examined the modern approach using the Mexican Intercensal Survey and remotely-sensed data available at the same time. Other types of passively-collected data (e.g., call detail records) or experiments using data from other countries could produce different results. Finally, there are other possible applications of machine learning to poverty mapping that we have not considered, such as extracting numerical covariates from high-dimensional data sources. Each of these limitations suggests a potential area for future research.

An additional area for future research would be to develop strategies to correct the R^2 for its various limitations. For example, one might correct for measurement error in the direct estimates by multiplying the estimated R^2 by the reciprocal of the bias term. This strategy would require estimating the variance of the measurement error associated with the direct estimates. Relatedly, note that our derivation of this bias relies on the assumption of classical measurement error, which may be violated in practice (e.g., if poorer or richer households are less likely to be sampled). Another topic for future research would then be to derive the bias under less stringent assumptions. Finally, as measurement error in the direct estimates is not the only concern with R^2 , future research could consider applying multiple adjustments simultaneously.

CRedit authorship contribution statement

Paul Corral: Writing – original draft, Validation, Supervision, Software, Methodology, Formal analysis, Conceptualization. **Heath Henderson:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Sandra Segovia:** Validation, Software, Formal analysis, Data curation.

Data availability

Data will be made available on request.

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³⁷ See Tzavidis et al. (2018) or Rao and Molina (2015) for further discussion of alternative validation procedures.

³⁸ In particular, see Hall et al. (2023) for detailed discussion on the capacity of remotely-sensed data to predict alternative welfare measures.

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